# CO1 PROJECT

## 4-BIT BINARY-TO-DECIMAL CONVERSION AND DISPLAY SYSTEM USING LOGIC GATES

# Introduction

Understanding binary input and decimal display

- Binary numbers use only two digits: 0 and 1.
- Digital systems naturally process information in binary form.
- A 4-bit input can represent values from 0 to 15.
- Converting the binary value for display makes the result easier for people to read.
- This project demonstrates the conversion and display process using digital logic.

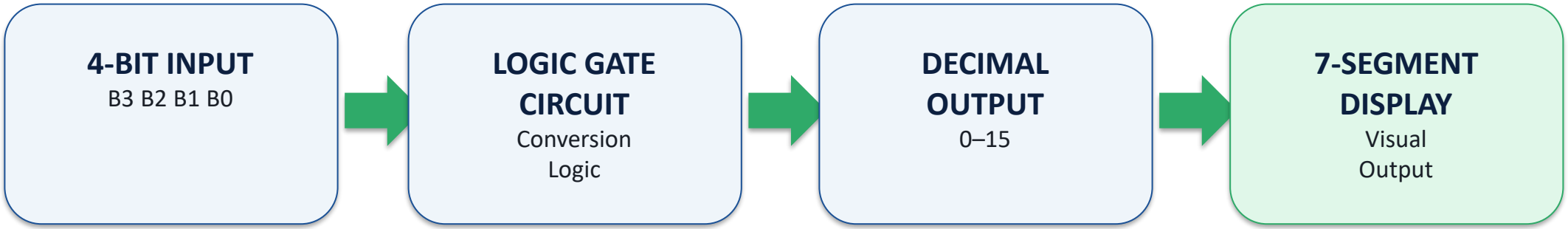
# Problem Statement & Objectives

**Problem: Design a digital circuit that accepts a 4-bit binary input and produces the corresponding decimal output for di**

- Accept 4-bit binary input using B3, B2, B1 and B0.
- Convert the input into its corresponding decimal value.
- Implement the required logic using gates.
- Display the result using a 7-segment display.
- Simulate and test the circuit in Logisim.

# Working Principle

Binary input is processed and then presented as a readable decimal display

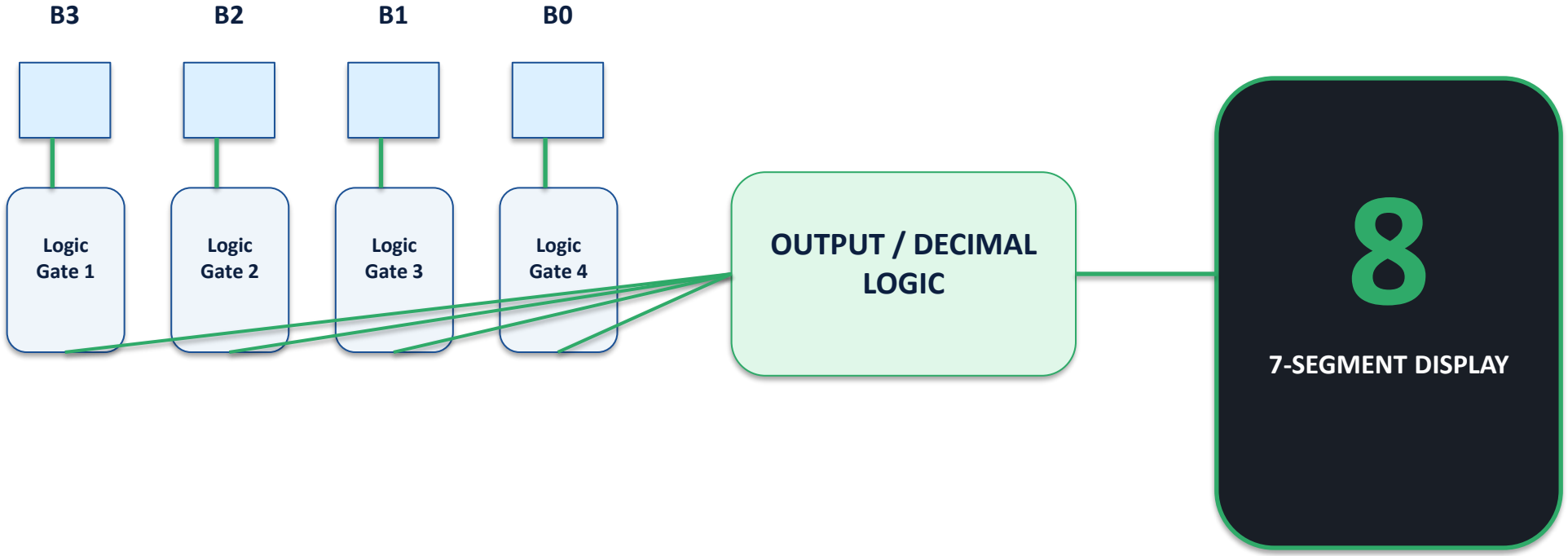



## Examples

0000 → 0 • 0001 → 1 • 0101 → 5 • 1001 → 9
1010 → 10 • 1011 → 11 • 1111 → 15

# Logic Circuit Design

Logisim-style implementation using four input switches, logic blocks and a 7-segment display




Circuit concept: B3–B0 are processed by combinational logic to generate the required display signals.

# Truth Table

All possible 4-bit inputs and their decimal values

| Binary Input | Decimal Value |
|---|---|
| 0000 | 0 |
| 0001 | 1 |
| 0010 | 2 |
| 0011 | 3 |
| 0100 | 4 |
| 0101 | 5 |
| 0110 | 6 |
| 0111 | 7 |
| 1000 | 8 |
| 1001 | 9 |
| 1010 | 10 |
| 1011 | 11 |
| 1100 | 12 |
| 1101 | 13 |
| 1110 | 14 |
| 1111 | 15 |

# Simulation & Results

Testing the circuit in Logisim

**Input: 0000**
Decimal: 0
**PASS**

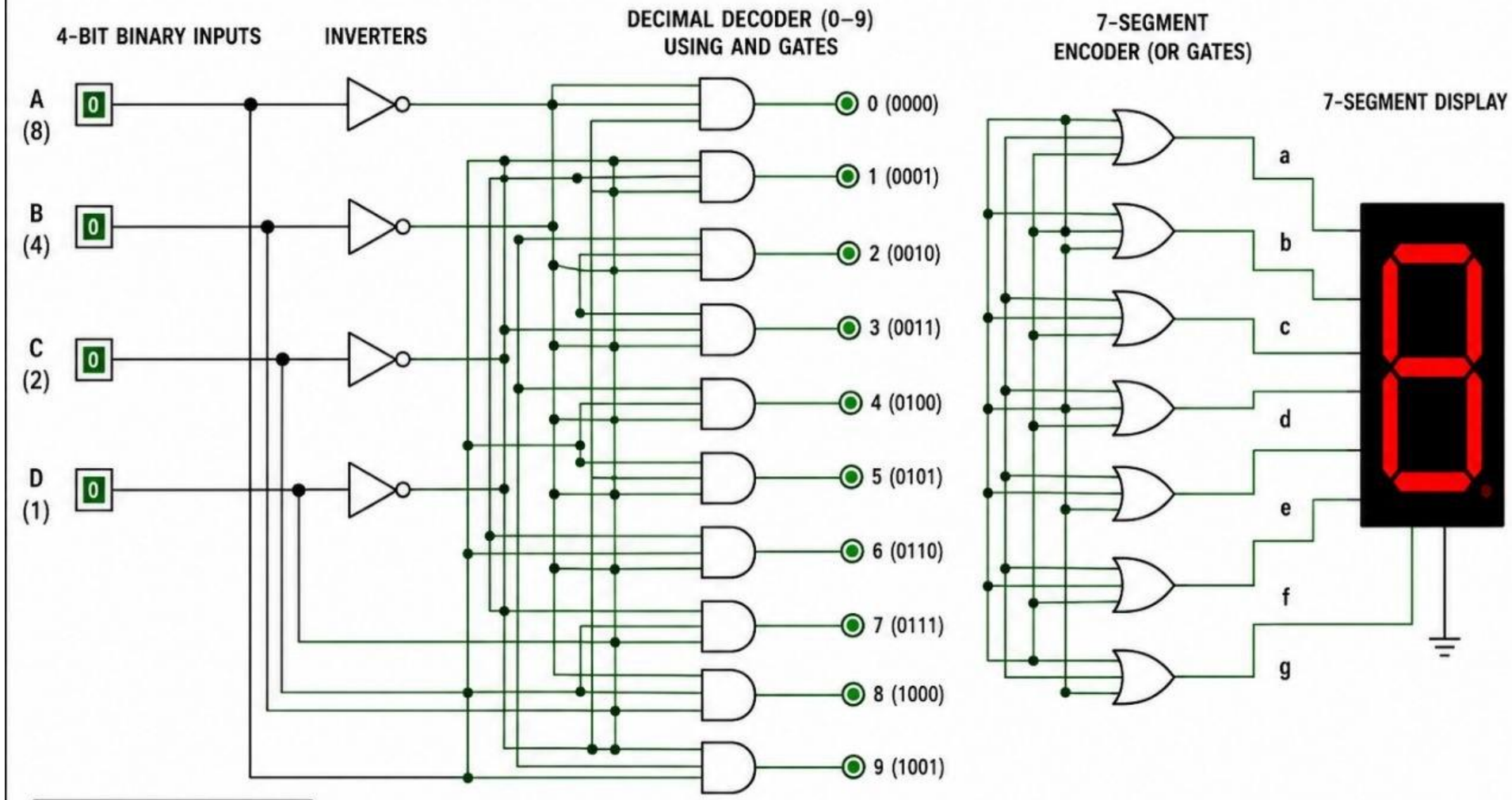
**Input: 0101**
Decimal: 5
**PASS**

**Input: 1001**
Decimal: 9
**PASS**

**Input: 1111**
Decimal: 15
**PASS**

- Binary input is provided through switches.
- Logic gates process the input combination.
- The resulting decimal value is represented on the display.
- Multiple input combinations are tested against expected values.

# 4-BIT BINARY-TO-DECIMAL CONVERSION AND DISPLAY SYSTEM USING LOGIC GATES




**INPUT ORDER (MSB → LSB)**

| A | B | C | D |
|---|---|---|---|
| (8) | (4) | (2) | (1) |

**OPERATION :**

→ 4-bit binary input (A B C D) is decoded to decimal (0–9)

→ The decimal value is displayed on 7-segment display

# CONCLUSION & FUTURE SCOPE

## Conclusion

- Designed a 4-bit binary conversion and display system.
- Demonstrated combinational logic and digital gates.
- Verified representative inputs through simulation.
- Provides an intuitive decimal representation of binary data.

## Future Scope

- Larger-bit binary inputs
- BCD conversion and multi-digit displays
- FPGA implementation
- Microcontroller integration
- Real-time digital display systems

# THANK YOU

E.YASWANTH2620030377
LOKESH-2620030368
GUNA NANDINI-2620030427
HAMISINI-2620030382